

Ben Irastorza

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EDUCATION

Georgia Institute of Technology

Dec 2026

Bachelor of Science in Computer Science

GPA: 3.6/4.0

- Relevant Coursework: Machine Learning, Systems and Networks, Database Systems, Computer Networking
- Involvement: CS/CompE Transfer Student Mentor, IFC Social Safety Coordinator, AXA Founding Member

EXPERIENCE

Software Engineering Intern

June 2026 – Present

Capital One

McLean, VA

- Working on the Prometheus Financial Core team to build and optimize critical cloud infrastructure.

Data Science Intern

Aug 2025 – Dec 2025

Southern Company

Birmingham, AL

- Developed the organization's most accurate winter peak demand forecast by implementing an XGBoost model with tailored meteorological features and data center load-growth analysis, reducing MAPE by 47%.
- Reduced data retrieval time by ~97% by developing a Python CLI tool leveraging the AVEVA PI Web API, replacing a manual Excel-based ingestion process with automated shell scripts for monthly analyses.
- Engineered a real-time AVEVA PI Vision dashboard measuring data center load swings' impact on Area Control Error (ACE) deviations, providing management the evidence needed to pursue new operational regulations.

Advanced Analytics Intern

May 2025 – Aug 2025

UCB Biopharma

Smyrna, GA

- Enabled identification of undiagnosed patients and risk stratification by building a BERT-based model that encodes longitudinal claims data into dense embeddings, supporting patient journey clustering across large populations.
- Strengthened risk assessment for an anti-seizure medication by developing a survival analysis model with left-truncation and right-censoring corrections, benchmarking internal outcomes against real-world evidence.
- Validated drug usage patterns for a rare pediatric epilepsy disorder by running subpopulation analyses on Komodo claims data in Snowflake, surfacing prescribing trends that informed commercial strategy.

Software Engineer Intern

May 2024 – Aug 2024

The Parker Avery Group

Atlanta, GA

- Accelerated executive decision-making by building a LangChain-based AI analytics agent that translates natural language questions into business metrics and automatically generates dashboards.
- Increased agent reliability by curating a human-validated question-to-SQL dataset and implementing Milvus semantic retrieval to surface similar queries, increasing accuracy by 16%.
- Scaled client-to-competitor product matching across thousands of SKUs by migrating to batched embeddings via the OpenAI Batch API, cutting embedding costs by 50%.

PROJECTS

Enhanced Spotify Wrapped | *Java, Android SDK, Firebase, GitHub Actions, Figma, Git*

- Built an Android app using the Spotify Web API that delivers personalized music recaps, tailored recommendations, and real-time AI-powered conversations with users' favorite artists.
- Automated build validation and deployment with a GitHub Actions CI/CD pipeline, reducing integration errors across a multi-sprint development cycle.
- Integrated Firebase Authentication with OAuth 2.0 to manage Spotify token exchange and session persistence, securing user credentials across sessions.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, HTML, CSS

Libraries/Frameworks: LangChain, Scikit-learn, XGBoost, PyTorch, Transformers, NumPy, Pandas, Matplotlib

Development Tools: Docker, Git, Jira, Figma, Wireshark, Power BI, Claude Code

Cloud/Databases: Databricks, Snowflake, PostgreSQL, MySQL, Firebase, Milvus, AVEVA PI System